SQL Project

Introduction:

This project focuses on analyzing purchase data from online learning platform sells various digital courses to learners across different countries using MYSQL. This platform offers various digital courses, and learners from different countries can purchase them.

The objective is to understand:

- Database creation
- Table relationships
- Constraints (Primary Key, Foreign Key)
- SQL Operations (DDL, DML, Queries)

Objectives:

- To create a structured database
  create database e_learnersdb;
- To establish relationships using foreign keys
- Understand learner behavior.
- Identify popular courses and categories.
- To apply SQL concepts like GROUP BY, HAVING, JOIN etc.

Database Design:

Table used:

1. **Courses.**

```
create table courses(
course_id int primary key,
courser_name varchar(100),
category varchar(50),
unit_price decimal(10,2)
);
```

2. **Purchases**

```
create table purchases (
purchase_id int primary key,
learner_id int,
course_id int,
quantity int,
```

```
purchase_date date,
foreign key (learner_id) references learners(learner_id),
foreign key (course_id) references courses(course_id)
);
```

3. **Learners**

```
create table learners(
learner_id int auto_increment primary key,
full_name varchar(50),
country varchar(50)
);
```

Table Structure:

Table: ***learners***

Attributes:

- *learner_id (Primary Key)*
- *full_name*
- Country

2. **Table:** ***courses***

Attributes:

- *course_id (Primary Key)*
- *course_name*
- *category*
- *unit_price*

3. **Table:** ***purchases***

Attributes:

- purchase_id (Primary Key)
- learner_id (Foreign Key → learners.learner_id)
- course_id (Foreign Key → courses.course_id)
- Quantity
- purchase_date

Relationships:

- ✓ One learner can purchase multiple courses
- ✓ Each courses belongs to one category
- ✓ Purchases table connects learners and courses.

Data Insertion:

Sample data was inserted:

-- Learners

```
insert into learners values
('1','Aarthi','India'),
('2','Shiv','India'),
('3','Udhay','USA'),
('4','Mary','UK'),
('5','Bommi','UAE'),
('6','Shri','Canada');
```

-- Courses

```
select *from courses;
select *from learners;
select *from purchases;
insert into courses values
(101, 'SQL','Database',2000.00),
(102,'Oracle','Database', 1000.00),
(103, 'Python', 'programming',800.00),
(104, ' C++','OOPS', 500.00),
(105, 'Data science','Analytics', 2500.00),
(106, ' Web developer', 'Programming', 1300.00);
```

--- purchases

```
insert into purchases values
(1,1, 101,1,'2024-01-10'),
(2,1,102,1,'2024-02-15'),
(3,2,103,2,'2024-03-1'),
(4,3,104,3,'2024-03-10'),
(5,4,101,2,'2024-04-05'),
(6,5,105,1,'2024-04-12'),
(7,2,102,2,'2024-05-1'),
(8,1,102,3,'2024-05-15');
```

SQL Queries and Analysis:

1. Total spending by Each learner

   Calculated total spending using:

   ```
   select
   l.full_name as learner_name,
   l.country,
   Round(p.quantity * c.unit_price,2) as Total_spending
   from purchases p
   join learners l on p.learner_id = l.learner_id
   join courses c on p.course_id = c.course_id
   order by Total_spending desc;
   ```

2. **Top 3 Most Purchased Courses:**

   ```
   select
   c.courser_name as Top_courses,
   (p.quantity) as Total_quantity_sold from purchases p
   join courses c on c.course_id = p.course_id
   order by Total_quantity_sold desc limit 3;
   ```

3. **Category-wise Revenue and Unique Learners**

   ```
   select
   c.category,
   round(sum(p.quantity*c.unit_price),2)as Total_revenue,
   count(distinct p.learner_id) as unique_learners from purchases p
   join courses c on c.course_id = p.course_id
   group by c.category
   order by total_revenue desc;
   ```

4. **Learners Purchasing Multiple Categories**

   ```
   select
   l.full_name as learners_name from purchases p
   join learners l on l.learner_id= p. learner_id
   join courses c on c. course_id = p.course_id
   group by l.full_name
   having count(distinct c.category)>1;
   ```

5. **Courses Not Purchased**

   ```
   select
   c.courser_name from courses c
   ```

```
left join purchases p on p.course_id = p.course_id
where p.course_id is null;
```

Data Presentation:

- ✓ Currency formatted using Roun(,2)
- ✓ Alias used for readability
- ✓ Result sorted using ORDERBY

Key Insights:

- ✓ Identified top spending learners
- ✓ Found most popular courses
- ✓ Analyzed revenue by category
- ✓ Detected courses with no sales

Conclusion:

This project demonstrates how sql can be used to analyze real-world data effectively. By using joins, aggregation and filtering, meaningful insights were derived from the dataset.